

Speaker notes: 10-minute presentation

Read alongside `slides/presentation\_10min.pptx`. The same notes are in each slide's notes pane.

Title (about 0:20)

- Our project asks one question: do carry trades crash when they are crowded?
 - We test it in two markets with the same economics: G10 currencies and crypto.
 - Strategy A is carry in seven G10 currencies against the US dollar, with a crowding filter built from CFTC data.
 - Strategy C is funding carry in twelve crypto perpetual futures on Binance, including the coins that collapsed, Terra and FTX Token.
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Slide 1. Carry pays because it crashes (about 1:00)

- Carry means borrowing in a low-yield asset and holding a high-yield one.
- Uncovered interest parity says this should earn nothing, but in the data it earns a premium.
- The catch is crash risk: carry goes up the stairs and down the elevator.
- Brunnermeier, Nagel and Pedersen show crashes are worst when speculators are crowded into the trade.
- In FX the yield is the interest-rate differential, and we measure crowding with CFTC speculator positions.
- Our FX universe is the yen, euro, British pound, Swiss franc, Canadian dollar, Australian dollar and New Zealand dollar, all against the US dollar.
- In crypto the yield is the perpetual funding rate, which also measures leveraged demand.
- Our coins are Bitcoin, Ethereum, BNB, XRP, Cardano, Dogecoin, Solana, Litecoin, Chainlink, Avalanche, Terra and FTX Token.
- We wrote the hypotheses and all parameters down before running any backtest.
- Uncovered interest parity (UIP) says a high-rate currency should fall by exactly the rate gap, so carry would earn zero; in practice high-rate currencies tend not to fall that much, which is the forward premium puzzle.
- 'Up the stairs and down the elevator' means carry earns small steady gains most of the time and then loses a lot in a few weeks, so its returns have negative skew.
- Crowding matters because when many speculators hold the same trade, a shock forces them all to unwind together, which makes the crash larger.
- Hypothesis A says three things: carry earns a premium, crowded months are followed by weaker carry, and carry loses when volatility spikes.

- Hypothesis C says crypto funding carry earns a premium with little exposure to Bitcoin, and that its crashes come from single coins.
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Slide 2. How the strategies trade (about 0:40)

- Here is how each strategy trades, step by step.
 - Strategy A trades once a month: at each month-end we rank the seven currencies by their interest rate against the US dollar, buy the top two and sell the bottom two.
 - Before trading we check crowding from CFTC data, which we use only after its Friday release; if crowding is above one, we halve the positions.
 - We hold for one month and then rank again.
 - Strategy C trades every Sunday: we rank the coins that are trading by their average funding over the past week, short the third with the highest funding and buy the third with the lowest, which is usually three coins on each side.
 - We hold for seven days, collecting funding on the shorts, and then rank again.
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Slide 3. Data and methodology (about 0:50)

- All data are public: FRED for FX and rates, CFTC for positions, Binance for crypto.
- One data fix worth mentioning: before 2000 the CFTC lists currency futures under a different exchange name; matching both names gave us positioning back to 1986.
- Every parameter was fixed in advance, and our out-of-sample period starts in 2011 for FX and in 2024 for crypto.
- Returns are net of costs, there is no look-ahead, and we keep the two coins that died, Terra (LUNA) and FTX Token (FTT).
- Out-of-sample means the later years that we did not look at while designing the strategy; if a rule still works there, it is less likely to be a result of data mining.
- Look-ahead bias means using information that was not public at the time of the trade; we avoid it because weights set at t earn the return of $t+1$, and CFTC positions measured on Tuesday are used only after their Friday release.
- Survivorship bias means testing only on assets that still exist today; we avoid it by keeping Terra and FTX Token, which together cost the crypto strategy about 100 percentage points of P&L.
- A basis point (bp) is 0.01%, so a cost of 3 bp means we lose 0.03% of every dollar we trade.
- The Lustig–Roussanov–Verdelhan (LRV) factor is the published academic carry factor, built by buying high-rate and selling low-rate currency portfolios; we use it only to check our own carry series.

Slide 4. Strategy A rule (about 1:00)

- Each month we rank the seven currencies by their three-month interest rate.
- We buy the two highest and sell the two lowest, so the portfolio has no net dollar bet.
- In practice we are almost always long the New Zealand and Australian dollars and short the Swiss franc, often with the yen or the euro.
- For crowding we take speculators' net futures position and z-score it over 36 months.
- Crowding is high when speculators are unusually long what we hold and short what we sell.
- When this score is above one standard deviation, we halve the position, which happened in 33 months.

Slide 5. Strategy A results (about 1:20)

- Plain carry earns about 3.3% a year after costs with 8.9% volatility, a net Sharpe of 0.37.
- Our carry has a 0.82 correlation with the published Lustig–Roussanov–Verdelhan factor, which validates the code and data.
- After crowded months, next-month carry averages only 0.02% versus 0.30% otherwise, with more negative skew.
- Out of sample the filter lifts the Sharpe from 0.27 to 0.32, and every variant in our robustness grid also beats 0.27.
- But it did not avoid 2008: the crash started from a non-crowded reading, so the max drawdown is -37% either way.
- One caveat is that there are only 33 crowded months, so the effect is not precisely estimated.

Slide 6. Strategy A risk (about 0:50)

- To see when it loses, we regress monthly carry on the dollar factor and on equity-volatility shocks.
- The volatility beta is -1.55 with a t-stat of -3.7: carry loses when volatility jumps.
- Once we control for these factors the alpha is not significant, so carry is pay for risk, not a free lunch.
- From July 2008 to March 2009 the strategy lost 20.5% as the funding currencies, the Japanese yen and the Swiss franc, rallied against the Australian and New Zealand dollars.

Slide 7. Strategy C (about 1:20)

- Crypto perpetual futures have their own interest rate, the funding rate, paid every 8 hours between longs and shorts.

- High funding means many leveraged longs, so it is both a carry signal and a crowding signal.
- Each Sunday we short the third of coins with the highest funding and buy the third with the lowest.
- The twelve coins are Bitcoin, Ethereum, BNB, XRP, Cardano, Dogecoin, Solana, Litecoin, Chainlink, Avalanche, Terra and FTX Token.
- We keep Terra (LUNA) and FTX Token (FTT), which both collapsed in 2022, to avoid survivorship bias.
- Net Sharpe is 0.14 over the full sample: -0.07 in 2020 to 2023 and 0.81 since 2024.
- After costs it earns 4.1% a year with 30% volatility, and its maximum drawdown is -67%, much deeper than FX carry.
- The worst week was the Terra (LUNA) collapse in May 2022, -27.3%: traders were shorting Terra, its funding turned negative, and so our rule held it long.
- A perpetual future has no expiry date, so the exchange uses the funding payment to keep its price close to the spot price.
- For example, if funding is +0.01% every 8 hours, longs pay shorts about 0.03% a day, which is roughly 11% a year.
- So shorting a coin with high funding earns that payment, just as selling a low-rate currency and buying a high-rate one earns the interest gap in FX.
- The book is dollar-neutral, which means the long and short sides have the same size, so a move of the whole crypto market should roughly cancel out.
- The out-of-sample Sharpe since 2024 is strong, but it comes from less than three years of data, so we treat it with caution.

Slide 8. Strategy C drivers (about 1:10)

- We split the return into funding and price. Funding adds about 8.5% a year, steadily.
- But price moves dominate: in 2020 to 2023 the coins we shorted, like Dogecoin and XRP, kept rallying and wiped out the funding.
- Since 2024 most of the gain came from price, so we read the strong out-of-sample number with caution.
- The beta to Bitcoin is about zero, so the strategy has no market exposure.
- Its losses come from single coins: Terra alone cost 87 percentage points, XRP 58 and Dogecoin 43, while Solana and BNB were the biggest winners.
- A risk control we fixed in advance made things worse and lowered the Sharpe to -0.16: the cap halved exposure and the filter missed the timing of Terra's collapse.

- Our takeaway is that crypto carry crashes in the mirror image of FX: it ends up long whatever traders short hardest.
 - As a robustness check we tried 6 lookback and rebalancing settings; all have a positive out-of-sample Sharpe, and the setting we fixed in advance is in the bottom half, so we did not pick the best one after the fact.
 - The table adds up: over the full sample, funding of +8.5%, price of -2.2% and costs of -2.2% give a net return of +4.1% a year.
 - In 2020 to 2023 funding paid 12.3% a year, but price moves took away 12.5%, so the strategy lost money.
 - Since 2024 funding paid only 3.1%, and price moves added 13.0%, which is luck rather than carry.
 - The risk control capped each coin at one sixth of its side of the book and refused to buy coins with 7-day funding below minus 50% a year.
 - It failed because the cap left part of the book in cash and halved the funding income, and on the Sunday before Terra collapsed its 7-day funding was only about minus 20% a year, so the filter did not remove it.
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Slide 9. Combined portfolio (about 0:50)

- Finally, we scale both strategies to 10% volatility and combine them 50/50.
- Their correlation is only 0.11: FX and crypto carry crash at different times.
- The mix has 7.0% volatility and a Sharpe of 0.51, and its worst month, -5.9%, is smaller than either part's.
- It does not beat FX alone on Sharpe, because crypto carry was weak in this window.
- Our takeaway is that the benefit of combining them is lower risk, not a higher return.
- We scale each strategy to 10% volatility so that neither one dominates the mix just because it is more volatile; crypto carry is about three times as volatile as FX carry before scaling.
- The scaling uses past volatility only, so there is no look-ahead in the combined portfolio.
- A correlation of 0.11 is close to zero, which means a bad month in one strategy says very little about the other.
- The window starts in 2021 because that is when both strategies have enough history for the volatility estimate.
- With only 67 months of overlap, these combined numbers are indicative rather than precise.

Slide 10. Conclusions (about 1:00)

- Our short answer is partly yes: crowding makes carry weaker, but it doesn't predict the biggest crashes.
- In FX, carry crashes are compensation for risk, and crowding is a modest warning sign, not a reliable crash predictor.
- In crypto, carry also earns a premium and also crashes, but the crashes come from single coins, mostly being long a coin that traders are crowding to short, so crowding tells you where the danger is rather than when it will hit.
- The table sums up our hypotheses: four are supported, crowding in FX leans yes, crypto funding carry is mixed, and our pre-set risk control did not work.
- In both markets, carry looks like compensation for crash risk.
- In FX, crowding measured from CFTC positions is a cheap and modestly useful warning sign.
- In crypto, sorting on funding alone exposes you to single-coin collapses like Terra and FTX Token.
- Because the two crash at different times, they diversify each other.
- What is new in our work is using CFTC positioning as a crowding filter on carry, and treating crypto funding as both the crypto interest-rate gap and a crowding signal, tested with the coins that died.
- The main limitations are a short crypto sample, few crowded FX months, and simple cost estimates.